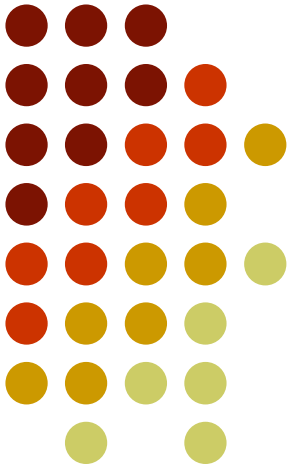
Interest Rate Futures

Chapter 6



Day Count Convention

- Defines:
  - the period of time to which the interest rate applies
  - The period of time used to calculate accrued interest (relevant when the instrument is bought or sold)

Day Count Conventions in the U.S.



Treasury Bonds:	Actual/Actual (in period)
Corporate Bonds:	30/360
Money Market Instruments:	Actual/360

Examples

- Bond: 8% Actual/ Actual in period.
  - 4% is earned between coupon payment dates. Accruals on an Actual basis. When coupons are paid on March 1 and Sept 1, how much interest is earned between March 1 and April 1?
- Bond: 8% 30/360
  - Assumes 30 days per month and 360 days per year. When coupons are paid on March 1 and Sept 1, how much interest is earned between March 1 and April 1?

Examples continued

- T-Bill: 8% Actual/360:
  - 8% is earned in 360 days. Accrual calculated by dividing the actual number of days in the period by 360. How much interest is earned between March 1 and April 1?

The February Effect (Business Snapshot 6.1, page 134)



- How many days of interest are earned between February 28, 2013 and March 1, 2013 when
  - day count is Actual/Actual in period?
  - day count is 30/360?

Treasury Bill Prices in the US

$$P = \frac{360}{n}(100 - Y)$$

Y is cash price per \$100

P is quoted price

Treasury Bond Price Quotes in the U.S

Cash price = Quoted price + Accrued interest since last coupon date

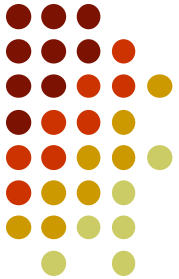
Quoted price: in dollars and thirty-seconds of a dollar for a bond with a face value of $100.

Suppose it is March 5, 2013, and the bond under consideration is an 11% coupon bond maturing on July 10, 2028, with a quoted price of 95-16 or $95.50. Coupons are paid semiannually. The most recent coupon date is January 10, 2013, and the next coupon date is July 10, 2013.

January 10, 2013 – March 5, 2013: 54 days

January 10, 2013 – July 10, 2013: 181 days

Treasury Bond Futures



Cash price received by party with short position =

Most Recent Settlement Price × Conversion factor + Accrued interest

Example

- Most recent settlement price = 90.00
- Conversion factor of bond delivered = 1.3800
- Accrued interest on bond =3.00
- Price received for bond is

$$1.3800 \times 90.00 + 3.00 = \$127.20$$

per $100 of principal

Conversion Factor



The conversion factor for a bond is approximately equal to the value of the bond on the assumption that the yield curve is flat at 6% with semiannual compounding

Example: Calculate Conversion Factor



- Consider a 10% coupon bond with 20 years and two months to maturity. For the purpose of calculation, the bond is assumed to have exactly 20 years to maturity.
- Consider an 8% coupon bond with 18 years and 4 months to maturity. For the purpose of calculation, the bond is assumed to have exactly 18 years and 3 months to maturity.

Cheapest-to-Deliver Bond



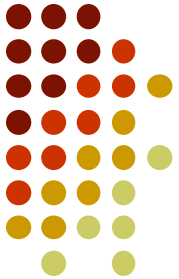
- At any given time during the delivery month, there are many bonds that can be delivered in the Treasury bond futures contract.
- The party with the short position can choose which of the available bonds is “cheapest” to deliver.
  - The party with the short position receives

  (Most recent settlement price × Conversion factor) + Accrued interest
  - The cost of purchasing a bond

  Quoted bond price + Accrued interest
  - The cheapest-to-deliver bond is the one for which

  Quoted bond price – (Most recent settlement price × Conversion factor) is least.

Example: Cheapest-to-Deliver Bond



- The party with the short position has decided to deliver and is trying to choose between the following three bonds. Assume that the most recent settlement price is 93-08, or 93.25.

Bond	Quoted bond price ($)	Conversion factor
1	99.50	1.0382
2	143.50	1.5188
3	119.75	1.2615

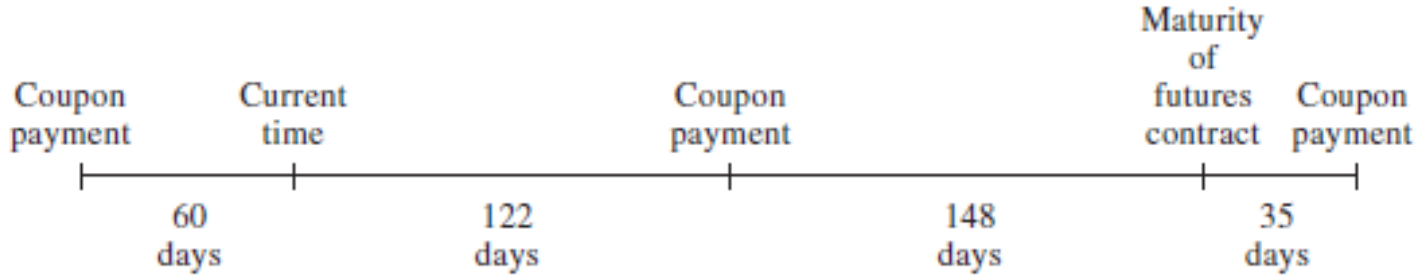
Determining the Futures Price

- If we assume that both the cheapest-to-deliver bond and the delivery date are known, the Treasury bond futures contract is a futures contract on a traded security that provides the holder with known income.

$$F_0 = (S_0 - I)e^{rT}$$

Example: Calculation of Treasury bond futures price

- Suppose that, in a Treasury bond futures contract, it is known that the cheapest-to-delivery bond will be a 12% coupon bond with a conversion factor of 1.6000. Suppose also that delivery will take place in 270 days. Coupons are payable semiannually on the bond. The last coupon date was 60 days ago, the next coupon date is in 122 days, and the coupon date thereafter is in 305 days. The term structure is flat, and the rate of interest (with continuous compounding) is 10% per annum. Assume that the current quoted bond price is $115.



Eurodollar Futures

- A Eurodollar is a dollar deposited in a bank outside the United States
- Eurodollar futures are futures on the 3-month Eurodollar deposit rate (same as 3-month LIBOR rate)
- One contract is on the rate earned on $1 million
- A change of one basis point or 0.01 in a Eurodollar futures quote corresponds to a contract price change of $25

Eurodollar Futures continued

- A Eurodollar futures contract is settled in cash
- When it expires (on the third Wednesday of the delivery month) the final settlement price is 100 minus the actual three month deposit rate

Example

Date	Quote
Nov 1	97.12
Nov 2	97.23
Nov 3	96.98
…….	…….
Dec 21	97.42

Example

- Suppose you buy (take a long position in) a contract on November 1
- The contract expires on December 21
- The prices are as shown
- How much do you gain or lose?

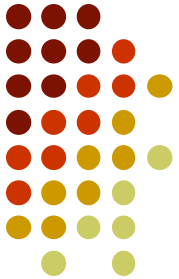
Example continued

- If on Nov. 1 you know that you will have $1 million to invest on for three months on Dec 21, the contract locks in a rate of

$$100 - 97.12 = 2.88\%$$

- In the example you earn 100 – 97.42 = 2.58% on $1 million for three months (=$6,450) and make a gain day by day on the futures contract of 30×$25 =$750

Formula for Contract Value (page 142)



- If Q is the quoted price of a Eurodollar futures contract, the value of one contract is

$$10{,}000[100-0.25(100-Q)]$$

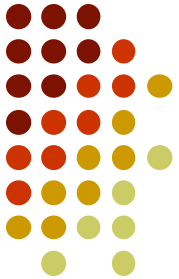
- This corresponds to the $25 per basis point rule

Example: Hedging with Eurodollar futures



- An investor wants to lock in the interest rate for a three-month period beginning September 16, 2015, on a principal of $100 million. We suppose that the September 2015 Eurodollar futures quote is 96.500, indicating that the investor can lock in an interest rate of 3.5% per annum. The investor hedges by buying 100 contracts.

Forward Rates and Eurodollar Futures (Page 143-144)

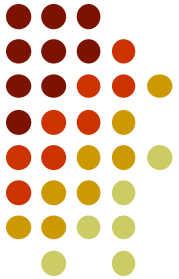


- Eurodollar futures contracts last as long as 10 years
- For short maturities (up to one year or so), the two contracts can be assumed to be the same and the futures rate can be assumed equal to the forward rate.
- For longer-dated contracts we cannot assume that the forward rate equals the futures rate

There are Two Reasons

- Futures is settled daily where forward is settled once
- Futures is settled at the beginning of the underlying three-month period; FRA is settled at the end of the underlying three-month period

Forward Rates and Eurodollar Futures continued



- A "convexity adjustment" often made is

  Forward Rate = Futures Rate $-0.5\sigma^2 T_1 T_2$

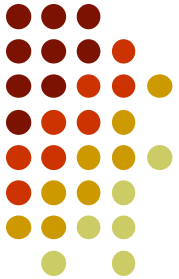
  - T_1 is the start of period covered by the forward/futures rate
  - T_2 is the end of period covered by the forward/futures rate (90 days later that T_1)
  - σ is the standard deviation of the change in the short-term rate per year

Convexity Adjustment when σ=0.012 (Table 6.3, page 145)



Maturity of Futures	Convexity Adjustment (bps)
2	3.2
4	12.2
6	27.0
8	47.5
10	73.8

Duration (page 144-148)



- Duration is a measure of how long on average the holder of the bond has to wait before receiving cash payments.
- Duration of a bond that provides cash flow c_i at time t_i is

$$D = \sum_{i=1}^{n} t_i \left[\frac{c_i e^{-yt_i}}{B} \right]$$

where B is its price and y is its yield (continuously compounded)

- This leads to

$$\frac{\Delta B}{B} = -D\Delta y$$

Example: Calculation of Duration

Time (years)	*Cash flow ($)*	*Present value*	*Weight*	*Time × weight*
0.5	5	4.709	0.050	0.025
1.0	5	4.435	0.047	0.047
1.5	5	4.176	0.044	0.066
2.0	5	3.933	0.042	0.083
2.5	5	3.704	0.039	0.098
3.0	105	73.256	0.778	2.333
Total:	130	94.213	1.000	2.653

Duration Continued

- When the yield y is expressed with compounding m times per year

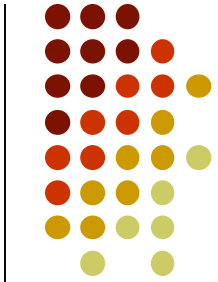
$$\Delta B = -\frac{BD\Delta y}{1+y/m}$$

- The expression

$$\frac{D}{1+y/m}$$

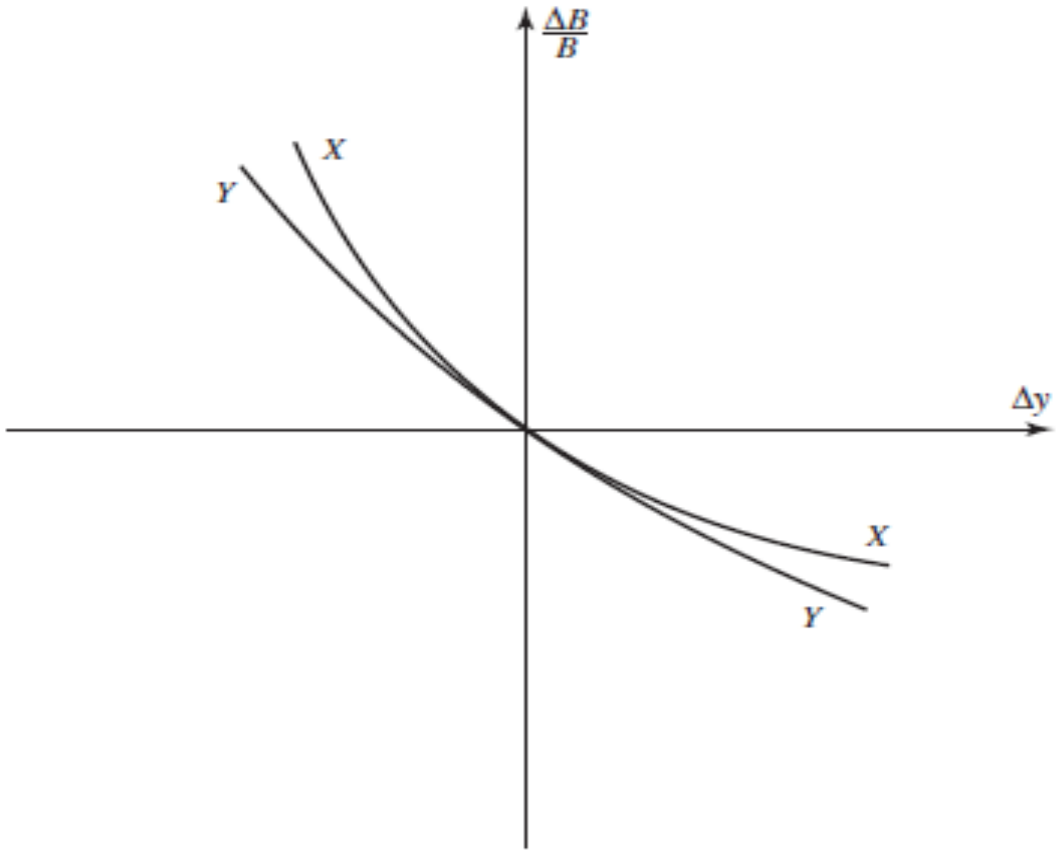
  is referred to as the “modified duration”

Bond Portfolios



- The duration of a bond portfolio can be defined as a weighted average of the durations of the individual bonds in the portfolio, with the weights being proportional to the bond prices.
- When duration is used for bond portfolio, there is an implicit assumption that the yield of all bonds will change by approximately the same amount – parallel shift
- The duration relationship applies only to small changes in yields.

Two Bond Portfolios with the Same Duration



Duration Matching (Portfolio Immunization)



- This involves hedging against interest rate risk by matching the durations of assets and liabilities
- It provides protection against small parallel shifts in the zero curve

Use of Eurodollar Futures



- One contract locks in an interest rate on $1 million for a future 3-month period
- How many contracts are necessary to lock in an interest rate on $1 million for a future six-month period?

Duration-Based Hedge Ratio

$$\frac{PD_P}{V_F D_F}$$

V_F Contract Price for Interest Rate Futures

D_F Duration of Asset Underlying Futures at Maturity

P Value of portfolio being Hedged

D_P Duration of Portfolio at Hedge Maturity

Example: Hedging a Bond Portfolio



- It is August 2. Three month hedge is required for a $10 million portfolio. Duration of the portfolio in 3 months will be 6.8 years.
- The quoted price for the December Treasury bond futures contract is 93-02 so that contract price is $93,062.50
- Duration of cheapest to deliver bond in 3 months is 9.2 years
- Number of contracts for a 3-month hedge is

$$\frac{10{,}000{,}000 \times 6.8}{93{,}062.50 \times 9.2} = 79.42$$

  - Short 79 December Treasury bond futures contracts on August 2
  - Close out the position on November 2

Example: Hedging a Floating-rate Loan



- It is April 29. A company has just borrowed $15 million for three months at an interest rate equal to one-month LIBOR plus 1% and would like to hedge its risk. The following quotes have been obtained:
  1. The one-month LIBOR rate is 8%.
  2. The June Eurodollar futures price is 91.88.
  3. The September Eurodollar futures price is 91.44.

  The company decides to take the following actions:
  1. Short five June contracts and five September contracts.
  2. Close out the June contracts on May 29.
  3. Close out the September contracts on June 29.

Limitations of Duration-Based Hedging



- Assumes that only parallel shift in yield curve take place
- Assumes that yield curve changes are small

Summary

- Day count convention
  - Actual/Actual, 30/360, Actual/360
- Treasury bill prices
  - Short-term: issued with terms of 4, 13, 26, or 52 weeks
  - $$P = \frac{360}{n}(100 - Y)$$
    Y is cash price per \$100
    P is quoted price

Summary

- Treasury bond price
  - Long-term: 20 or 30 years
  - Cash price = Quoted price + Accrued interest since last coupon date
- Treasury bond futures
  - Cash price received by party with short position = Most Recent Settlement Price × Conversion factor + Accrued interest
  - Conversion factor calculation
    - Rounded down to the nearest 3 months for purpose of calculation
    - Case 1: 20 years to maturity
    - Case 2: 18 years and 3 months to maturity

Summary

- Treasury bond futures (cont'd)
  - Cheapest-to-deliver bond
    - Quoted bond price – (Most recent settlement price × Conversion factor) is least.
  - Futures price
    - $F_0 = (S_0 - I)e^{rT}$
    - Accrued interest calculation

Summary

- Eurodollar futures
  - Futures on the 3-month Eurodollar deposit rate (same as 3-month LIBOR rate)
  - Contract size = $1 million
  - One basis point change (0.01%) corresponds to a change of $25
  - Settled in cash
  - Quoted price = 100 – actual 3-month deposit rate
  - Contract value = 10,000[100-0.25(100-Q)]
  - FRA and Eurodollar futures
    - Convexity adjustment: Forward Rate = Futures Rate$-0.5\sigma^2 T_1 T_2$

Summary

- Duration
  - A measure of how long on average the holder of the bond has to wait before receiving cash payments
    - Continuous compounding

$$D = \sum_{i=1}^{n} t_i \left[\frac{c_i e^{-yt_i}}{B} \right] \qquad \frac{\Delta B}{B} = -D\Delta y$$

    - Compounding *m* times per year

$$\Delta B = -\frac{BD\Delta y}{1 + y/m} \qquad D^* = \frac{D}{1+y/m}$$ (modified duration)

    - The duration relationship applies only to small changes in yields.

Summary

- Duration-based hedging
  - Hedging against interest rate by matching the durations of assets and liabilities
  - Hedge ratio

$$\frac{PD_P}{V_F D_F}$$

  - Limitations
    - Parallel shift and small changes in yields